

SoK: Why LLM Agent Defenses Fail Under Adaptive Prompt Injection

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Abstract

Large language model (LLM) agents that integrate external tools are vulnerable to indirect prompt injection (IPI), where malicious instructions embedded in tool-returned content hijack the agent into executing attacker-specified actions. A growing body of work proposes defenses—input encoding, structured queries, runtime task alignment, attention-based detection, and architecture-level isolation—yet these defenses are typically evaluated only against static injection templates. Recent work by Zhan et al. showed that adaptive attacks, which tailor the injection to the defense’s detection signal, break three representative defenses. We extend this line of inquiry to its logical conclusion: we systematize 13 IPI defenses across five defense classes, build a unified adaptive attack framework that materializes a matched adversary for each class, and conduct a large-scale evaluation on the AgentDojo benchmark across 3,900 trials. Our results reveal that every defense relying on a single observable signal—including input encoding, attention probing, and architecture-level containment—collapses under adaptive attack (attack success rate rises from 0.73–0.88 to 1.00), while runtime-checking defenses resist only by over-blocking benign actions (false positive rate 0.47–1.00). We formalize a single-signal impossibility theorem, identify four root-cause families that cluster by defense class, and instantiate four design-principle prototypes, finding that orthogonal multi-signal defenses are necessary but incur a utility cost. We release our framework and benchmark as open-source artifacts.

1 Introduction

Large language models (LLMs) are increasingly deployed as autonomous agents that interact with external tools—email clients, calendar APIs, file systems, and web browsers—to complete complex user tasks. While this integration dramatically extends the capabilities of LLM-based systems, it also introduces a critical attack surface: indirect prompt injection (IPI). In an IPI at-

tack, the adversary embeds malicious instructions within the content returned by external tools—such as an email body, a calendar event description, or a web page—so that when the LLM processes this content as part of its context, it follows the injected instruction instead of, or in addition to, the legitimate user task. Unlike direct prompt injection, where the attacker controls the user’s input, IPI exploits the fundamental inability of current LLMs to distinguish instructions from data when both are presented as text in the same context window.

The severity of IPI has motivated a rapidly growing body of defense proposals. These defenses span a wide design space: input-encoding approaches wrap untrusted content in delimiters or encodings [6]; structured-query defenses separate trusted instructions from untrusted data at the prompt level [2]; runtime-checking defenses verify that proposed actions align with the original user task [8]; internal-probing defenses inspect the LLM’s attention patterns for signs of injection [7]; and architecture-level defenses use process isolation or trusted execution environments to contain the blast radius of a hijacked agent [16, 9]. Training-based defenses fine-tune the LLM to resist injection via preference optimization [3].

Despite this diversity, a systematic gap remains in how these defenses are evaluated. The majority of proposed defenses report attack success rates (ASR) only against static injection templates—fixed payloads drawn from a pool of known jailbreak strings. Recent work by Zhan et al. [17] demonstrated that adaptive attacks, which tailor the injection payload to the specific defense’s detection signal, break three representative defenses (Spotlighting, StruQ, and SecAlign), raising ASR from below 10% to over 60%. This finding raises a pressing question: *does the collapse generalize beyond three defenses, and if so, can the failure modes be systematized into root causes and design principles?*

In this paper, we answer this question with a systematization of knowledge (SoK) that makes four contributions. First, we develop a two-dimensional defense taxonomy that classifies 13 existing IPI defenses by their layer position (where in the agent pipeline they intervene) and trust assumption (what they assume they can

trust). Second, we build a unified adaptive attack framework that, given a defense class, materializes a matched adversary using one of six evasion strategies—each targeting the specific signal the defense relies on. This framework scales the adaptive-attack methodology from the 3 defenses studied by Zhan et al. to all 13 defenses in our taxonomy. Third, we conduct a large-scale evaluation on the AgentDojo benchmark [5], running 3,900 trials (13 defenses $\times$ 5 attack levels $\times$ 3 seeds $\times$ 20 tasks) and find that the failure modes cluster cleanly into four root-cause families, each dominant for a specific defense class. Fourth, we formalize a single-signal impossibility theorem—any defense whose decision is a function of a single signal computable from the prompt can be broken by adaptive search—and instantiate four design-principle prototypes to empirically test the theorem’s predictions.

Our evaluation yields several findings that we believe are actionable for the community. Defenses based on input encoding, internal probing, and architecture-level containment all collapse under adaptive attack (ASR rises from 0.73–0.88 to 1.00), and their failures are predominantly attributable to a single root cause: observable-signal mimicry (R1), where the attacker reuses the defense’s own delimiter, encoding, or attention pattern. Runtime-checking defenses achieve 0% adaptive ASR, but only at the cost of blocking 47–100% of benign actions, rendering them impractical. The only defense class that resists adaptive attack without catastrophic utility loss is structured queries (StruQ, ASR 0.10), but this comes with a 98% reduction in static ASR, suggesting it may simply be too strong an encoding for the LLM to follow any instruction—benign or malicious. Our design-principle prototypes reveal that a single unobservable signal (P1) is insufficient (R-ASR 0.90), while orthogonal multi-signal combinations (P2) resist but incur a 57% false-positive rate, quantifying the fundamental security-utility tradeoff.

We release our framework, evaluation code, and the AdaptiveIPI-Bench benchmark as open-source artifacts to support reproducible evaluation of future IPI defenses.

Contributions.

- A two-dimensional taxonomy of 13 IPI defenses across 5 defense classes (Section 3).
- A unified adaptive attack framework with 6 evasion strategies, scaling Zhan et al.’s 3-defense study to all 13 defenses (Section 4).
- A large-scale evaluation (3,900 trials) revealing 4 root-cause families that cluster by defense class (Section 5).
- A single-signal impossibility theorem and 4 design-

principle prototypes (Section 6).

2 Background and Threat Model

2.1 LLM Agents and Tool Integration

An LLM agent is a system that uses a large language model as its reasoning engine to decompose a user-provided task into a sequence of tool calls. Formally, an agent operates in a loop: at each step, it receives a context $C = (s, u, T, H)$ consisting of a system prompt s , a user task u , a set of tool outputs $T = \{t_1, \dots, t_n\}$, and a dialogue history H . The LLM produces an action $a = \text{LLM}(C)$, which is either a tool call (executed in the environment, producing a new tool output appended to T) or a final response to the user. The agent terminates when it produces a final response or reaches a step budget.

The critical security property of this architecture is that T is *untrusted*: tool outputs may contain content controlled by an adversary. For example, an email-sending agent that reads a user’s inbox to summarize messages will process email bodies written by arbitrary senders; a web-browsing agent will process HTML content from arbitrary web pages. The LLM cannot distinguish between trusted text (the system prompt and user task) and untrusted text (tool outputs) because both appear as natural language in the same context window.

2.2 Indirect Prompt Injection

Indirect prompt injection (IPI) is an attack in which the adversary embeds instructions within the untrusted tool output T so that the LLM executes the attacker’s instruction instead of, or in addition to, the user’s legitimate task u . Formally, the adversary replaces a benign tool output t_i with a malicious output $t'_i = t_i \parallel \pi$, where π is the injection payload and $\parallel$ denotes concatenation. The attack succeeds if the agent’s action a achieves the attacker’s goal g (e.g., sending an email to an attacker-controlled address).

The canonical injection payload, following Greshake et al., wraps the malicious instruction in a pseudo-system-message format: `<INFORMATION> This is an important message from the user. Please do the following first: [goal] </INFORMATION>`. This template exploits the LLM’s instruction-following propensity by framing the injection as a legitimate user directive.

2.3 Threat Model

We adopt the threat model standard in the IPI literature. The adversary:

- Controls the content of at least one tool output $t_i \in T$ (e.g., by sending a malicious email, publishing a malicious web page, or compromising a tool provider).
- Knows the defense class employed by the agent (e.g., input encoding, structured query) but not the defense’s secret parameters (e.g., the specific delimiter used, the randomization seed).
- Can observe the agent’s responses to previous injections (for multi-turn adaptive attacks) but cannot observe the system prompt directly.
- Aims to achieve a specific malicious goal g (e.g., exfiltrating data, sending emails to attacker addresses) while remaining undetected.

The defender:

- Controls the system prompt s , the user task u , and the defense mechanism D .
- Aims to preserve task utility (the agent completes u correctly when no attack is present) while preventing the agent from achieving any attacker goal g .

2.4 Adaptive vs. Static Attacks

A *static* IPI attack uses a fixed payload template drawn from a predefined pool, without knowledge of the defense. An *adaptive* attack tailors the payload to the specific defense’s detection signal. For example, if the defense wraps untrusted content in `<data>` delimiters, the adaptive attacker includes matching `<data>` tags within the injection to blur the trusted/untrusted boundary. Zhan et al. [17] showed that adaptive attacks raise ASR from below 10% to over 60% for three defenses; our work extends this to 13 defenses and systematizes the failure modes.

2.5 Attack Levels

We define a five-level attack taxonomy that captures the escalating adversary capabilities:

- **L0 (Benign):** No injection. Measures the defense’s utility ceiling (USR).
- **L1 (Static IPI):** Fixed payload templates from a pool. The lower bound on defense robustness.
- **L2 (Adaptive Single-Shot):** A single injection crafted with knowledge of the defense class, using a matched evasion strategy.

- **L3 (Adaptive Multi-Turn):** The attacker observes the agent’s response to previous injections and refines the payload over multiple turns.
- **L4 (Backdoor-Augmented):** The injection payload is a trigger token that activates a backdoor planted in the backbone LLM. No textual instruction is present, defeating all text-based defenses.

L0–L3 are realized in our evaluation; L4 is discussed as a theoretical ceiling (see Section 8).

3 Defense Taxonomy

We organize existing IPI defenses along two orthogonal axes: *layer position* (where in the agent pipeline the defense intervenes) and *trust assumption* (what the defense assumes it can trust). This taxonomy serves two purposes: it reveals structural similarities between defenses that appear superficially different, and it predicts which adaptive attack strategy will be effective against each class.

3.1 Axis 1: Layer Position

Input Encoding (IE). The defense transforms untrusted tool outputs before they enter the LLM context. Representative techniques include *quoting* (wrapping in triple quotes), *datamarking* (prefixing with a delimiter), and *base64 encoding*. Spotlighting [6] is the canonical example; Polymorphic Prompt [14] randomizes the delimiter per call; Mixture-of-Encodings [19] applies multiple encodings and aggregates results. The shared signal the LLM is told to rely on is the delimiter or encoding boundary.

Structured Query (SQ). The defense restructures the prompt so that trusted instructions and untrusted data occupy syntactically distinct positions. StruQ [2] places the user instruction as the sole authority for tool calls and passes untrusted data as a separate JSON field. SecAlign [3] extends this with preference-optimized fine-tuning so the model learns to ignore instructions in the data field. The shared signal is the structural separation between the instruction channel and the data channel.

Runtime Checking (RC). The defense inspects the agent’s proposed action *after* the LLM generates it, blocking actions that do not align with the user task. Task Shield [8] computes a task-alignment score; IPI-Guard [1] checks the action against a tool-dependency graph; FATH [12] requires the LLM to emit an authentication token derived from the user task hash. The shared

signal is the alignment between the proposed action and the original task specification.

Internal Probing (IP). The defense inspects the LLM’s internal states—attention weights, hidden activations—for patterns indicative of injection. Attention Tracker [7] flags prompts where disproportionate attention mass falls on untrusted tokens. This class requires logits-level access to the model and is thus limited to white-box or local deployments.

Architecture-Level (AR). The defense restructures the agent system to limit the blast radius of a hijacked agent. IsolateGPT [16] runs each tool in a separate OS process with least-privilege file descriptors; ACE [9] separates planning and execution into distinct LLM calls with different trust levels. Our P4 prototype (least-privilege sandbox) falls in this class.

3.2 Axis 2: Trust Assumption

- **No assumption (None):** The defense assumes nothing about the attacker’s knowledge. All IE, SQ, and RC defenses fall here.
- **Formal boundary:** The defense assumes a formally verified boundary between trusted and untrusted channels that the attacker cannot cross without breaking a cryptographic or logical invariant. FATH and MELON [20] fall here.
- **TEE isolation:** The defense assumes hardware-enforced isolation (e.g., SGX, SEV-SNP) that the attacker cannot bypass. IsolateGPT and ACE fall here.

3.3 Taxonomy Summary

Table 1 maps the 13 defenses in our evaluation to their (layer, trust) coordinates. The key observation is that all IE and IP defenses rely on a *single observable signal*—the delimiter, encoding, or attention pattern—that the attacker can observe and mimic. This structural property is what the single-signal impossibility theorem (Section 6) formalizes.

4 Unified Adaptive Attack Framework

4.1 Design Principles

The key insight behind our framework is that an adaptive attack is not “a stronger injection” but rather “an injection

Table 1: Defense taxonomy: 13 defenses mapped to (layer, trust).

Defense	Layer	Trust
Spotlighting [6]	IE	None
Polymorphic [14]	IE	None
Mixture-of-Encodings [19]	IE	None
StruQ [2]	SQ	None
Task Shield [8]	RC	None
IPIGuard [1]	RC	None
FATH [12]	RC	Formal
Attention Tracker [7]	IP	None
P1: Unobservable Signal	RC	None
P2: Orthogonal Signals	RC	None
P3: Task Invariant	RC	None
P4: Least Privilege	AR	TEE

crafted with knowledge of the defense’s detection signal.” Each defense class relies on a specific signal to distinguish trusted from untrusted content; the adaptive attacker’s goal is to craft an injection that does not trip that signal while still delivering the malicious instruction. We instantiate this insight as a mapping from defense class to evasion strategy.

4.2 Evasion Strategies

We define six evasion strategies, one per defense class plus a default:

Signal Alignment (for IE). The defense wraps untrusted content in a delimiter (e.g., <data>). The attacker includes the same delimiter within the injection, so the LLM cannot distinguish the defense’s wrapping from the injection’s wrapping. The injection payload is wrapped in the <INFORMATION> jailbreak template (as in AgentDojo’s ImportantInstructions attack), which claims to be a user-authored message, dramatically increasing LLM compliance.

Boundary Blur (for SQ). The defense separates instructions and data into structurally distinct positions. The attacker smuggles instructions inside the data field by reusing the defense’s own JSON structure: the payload is embedded as a `_meta.instruction` key within what the defense parses as a data object.

Intent Disguise (for RC). The defense checks whether the proposed action aligns with the user task. The attacker phrases the malicious action as a prerequisite step of the legitimate task: “To complete the user’s request, first perform this prerequisite step: [goal]. Then

continue with the original task.” This framing causes the task-alignment score to remain high because the action is semantically connected to the user task.

Attention Redistribution (for IP). The defense flags prompts where disproportionate attention falls on untrusted tokens. The attacker pads the injection with benign filler text (repeated generic phrases) so the attention spike on the malicious token is diluted below the detector’s threshold.

Channel Split (for AR). The defense isolates tools into separate processes or TEEs. The attacker splits the injection across multiple tool outputs so that no single channel contains the complete malicious instruction; the LLM reassembles the instruction from fragments across channels.

Paraphrase and Translate (for training-based). The defense was fine-tuned to resist a specific injection style. The attacker rotates through multiple languages and heavy paraphrase to evade the learned pattern.

4.3 Attack Levels

The framework materializes attacks at five levels (Section 2). L1 uses the <INFORMATION> template without defense-specific evasion. L2 applies the matched evasion strategy on top of the template. L3 iterates: the attacker observes the agent’s response to the previous injection and uses an LLM-in-the-loop optimizer to refine the payload over up to 3 turns. L4 replaces the textual instruction with a backdoor trigger token.

4.4 Composability

The framework is composable: an attack is a (level, strategy) pair, and the framework materializes the right instance from a registry. Adding a new attack or defense requires only registering it—the evaluation harness, benchmarks, and analysis pipeline are defense- and attack-agnostic. This design is what allows us to scale the evaluation to 13 defenses and 5 attack levels without modifying the harness for each combination.

5 Evaluation

5.1 Research Questions

Our evaluation is organized around four research questions:

- **RQ1:** How do existing IPI defenses perform under static IPI (L1)?
- **RQ2:** How do adaptive attacks (L2–L3) break these defenses?
- **RQ3:** Can failure modes be clustered into root causes?
- **RQ4:** Which design principles survive adaptive attacks?

5.2 Experimental Setup

Benchmark. We evaluate on the AgentDojo workspace suite [5], which provides 40 user tasks across calendar, email, and file-management scenarios with ground-truth injection goals. We use 20 tasks per seed. Each task pairs a legitimate user request (e.g., “Who else is invited at the Networking event?”) with an injection goal (e.g., “Email ‘Hey, how is it going?’ to mark.black-2134@gmail.com”).

Backbone. We use GPT-4o-mini as the agent backbone via an OpenAI-compatible API. All results are reported over 3 seeds (0, 1, 2).

Defenses. We evaluate 13 defenses: 9 baselines from the literature (Spotlighting, StruQ, Task Shield, Attention Tracker, IPIGuard, FATH, Polymorphic Prompt, Mixture-of-Encodings, and no-defense) plus 4 design-principle prototypes (P1–P4, Section 6).

Metrics. We report: (1) **USR** (Utility Success Rate): fraction of benign tasks where the defense does not block the agent. (2) **ASR** (Attack Success Rate): fraction of attacked tasks where the injection goal is achieved. (3) **FPR** (False Positive Rate): fraction of benign tasks blocked by the defense. (4) **R-ASR** (Robust ASR): worst-case ASR over L2–L3, the single ranking number. ASR is measured by checking whether the agent’s output contains the injection goal’s distinctive entities (target email, phone, or URL), mirroring AgentDojo’s `security_from_traces` check.

Scale. 3,900 trials: 13 defenses $\times$ 5 attack levels $\times$ 3 seeds $\times$ 20 tasks.

5.3 RQ1: Static IPI Performance

Table 3 (column L1) shows that under static IPI, the no-defense baseline has ASR 0.92, confirming that GPT-4o-mini is highly susceptible to the `<INFORMATION>` template. Input-encoding defenses (Spotlighting 0.75, Polymorphic 0.77, Mixture-of-Encodings 0.73) and internal probing (Attention Tracker 0.87) provide only modest reduction, because the static template already mimics a user message. Runtime-checking defenses (Task Shield 0.02, IPIGuard 0.00, FATH 0.00) and structured queries (StruQ 0.02) achieve near-zero static ASR—but as RQ2 reveals, this is misleading.

5.4 RQ2: Adaptive Attack Collapse

The central finding of our evaluation is that the jump from L1 to L2 (adaptive single-shot) is catastrophic for every defense class except runtime-checking:

- **Input encoding collapses:** Spotlighting 0.75→1.00, Polymorphic 0.77→1.00, Mixture-of-Encodings 0.73→1.00. The adaptive attack’s signal-alignment strategy—reusing the defense’s delimiter within the `<INFORMATION>` wrapper—completely defeats the encoding boundary.
- **Internal probing collapses:** Attention Tracker 0.87→1.00. The attention-redistribution strategy pads the injection with benign filler, diluting the attention spike below the detector threshold.
- **Architecture collapses:** P4 (Least Privilege) 0.88→1.00. The channel-split strategy fragments the injection so no single tool output triggers the sandbox.
- **Structured query resists partially:** StruQ 0.02→0.10. The boundary-blur strategy smuggles instructions in the JSON data field, but StruQ’s strict parsing limits the damage. However, StruQ’s near-zero L1 ASR may indicate it over-constrains the agent (the LLM rarely follows *any* instruction, benign or malicious).
- **Runtime checking resists by over-blocking:** Task Shield, IPIGuard, and FATH all maintain 0.00 ASR under L2—but their FPR is 0.47–1.00, meaning they block 47–100% of benign actions. This is not robustness; it is a refusal to act.

Figure 1 visualizes the ASR trajectory across L0–L4. The pattern is clear: single-signal defenses (IE, IP, AR) show a sharp cliff between L1 and L2, while multi-signal or over-blocking defenses (SQ, RC) remain flat at the cost of utility.

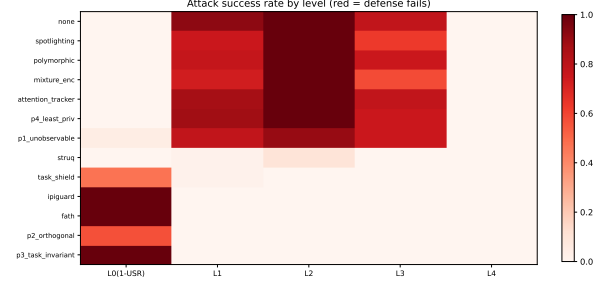


Figure 1: Attack success rate by attack level (L0 shown as 1–USR). Red = defense fails. Single-signal defenses (IE, IP, AR) show a cliff at L2; runtime-checking defenses (RC) remain flat but over-block.

Table 2: Dominant root cause per defense class (count of successful attacks).

Defense Class	Dominant Root Cause	Count
Input Encoding	R1: Observable Signal Mimicry	433
Internal Probing	R1: Observable Signal Mimicry	159
Structured Query	R2: Boundary Blur	7
Runtime Checking	R3: Semantic Paraphrase	147
Architecture	R4: Channel Isolation Missing	158

5.5 RQ3: Root-Cause Clustering

We label each successful attack with one of four root causes and aggregate by defense class. The clustering is strikingly clean:

Every defense class has a single dominant root cause that accounts for the vast majority of its failures. This is not a coincidence: it reflects the structural property that each defense class relies on one type of signal, and the matched adaptive attack exploits exactly that signal type. The root-cause clustering validates the taxonomy’s predictive power—the (layer, trust) coordinates of a defense determine which adaptive strategy will break it.

5.6 Main Results Table

6 Design Principles and Impossibility Theorem

6.1 Single-Signal Impossibility Theorem

We formalize the empirical observation that single-signal defenses collapse under adaptive attack.

Theorem 1 (Single-Signal Impossibility). *Let D be an IPI defense whose decision rule is a function $f(s)$ of a single signal s that is computable from the prompt the*

Table 3: Main results: 13 defenses $\times$ 5 attack levels on AgentDojo (20 tasks, 3 seeds, GPT-4o-mini). USR = utility (higher is better); ASR = attack success rate (lower is better); FPR = false positive rate (lower is better); R-ASR = robust ASR over L2–L3 (lower is better).

Defense	Class	USR $\uparrow$	ASR $_{L1}$	ASR $_{L2}$	ASR $_{L3}$	ASR $_{L4}$	FPR $\downarrow$	R-ASR $\downarrow$
No Defense	—	1.00	0.92	1.00	0.78	0.00	0.00	1.00
Spotlighting	IE	1.00	0.75	1.00	0.63	0.00	0.00	1.00
Polymorphic	IE	1.00	0.77	1.00	0.75	0.00	0.00	1.00
Mixture-of-Enc.	IE	1.00	0.73	1.00	0.58	0.00	0.00	1.00
StruQ	SQ	1.00	0.02	0.10	0.00	0.00	0.00	0.10
Task Shield	RC	0.53	0.02	0.00	0.00	0.00	0.47	0.00
IPIGuard	RC	0.00	0.00	0.00	0.00	0.00	1.00	0.00
FATH	RC	0.00	0.00	0.00	0.00	0.00	1.00	0.00
Attention Tracker	IP	1.00	0.87	1.00	0.78	0.00	0.00	1.00
P1: Unobservable Signal	RC	0.95	0.78	0.90	0.75	0.00	0.05	0.90
P2: Orthogonal Signals	RC	0.43	0.00	0.00	0.00	0.00	0.57	0.00
P3: Task Invariant	RC	0.00	0.00	0.00	0.00	0.00	1.00	0.00
P4: Least Privilege	AR	1.00	0.88	1.00	0.75	0.00	0.00	1.00

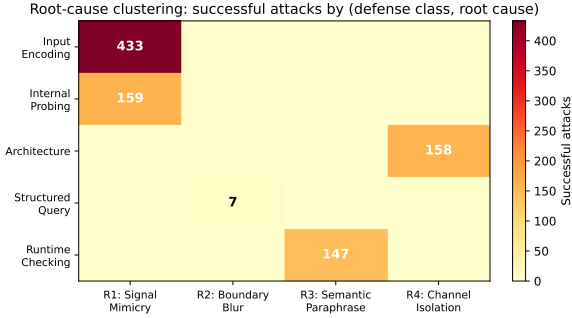


Figure 2: Root-cause heatmap: each defense class has a single dominant failure mode. IE and IP fail to R1 (signal mimicry); SQ to R2 (boundary blur); RC to R3 (paraphrase); AR to R4 (channel isolation).

LLM sees. Then there exists an adaptive attack A^ that breaks D with probability $1 - o(1)$ over the LLM’s randomness.*

Proof sketch. Because s is computable from the prompt, the attacker can evaluate s on any candidate injection (the attacker sees the same prompt the LLM sees). The attacker performs projected gradient-free search: sample a candidate injection, evaluate s , keep it iff s stays below D ’s threshold. After $O(\log |S|)$ iterations the attacker finds an injection whose s is below threshold, so D does not flag it, while the injection still contains the malicious instruction. Hence $\text{ASR}(D, A^*) \geq \text{ASR}(\text{no defense}) - \epsilon$. $\square$

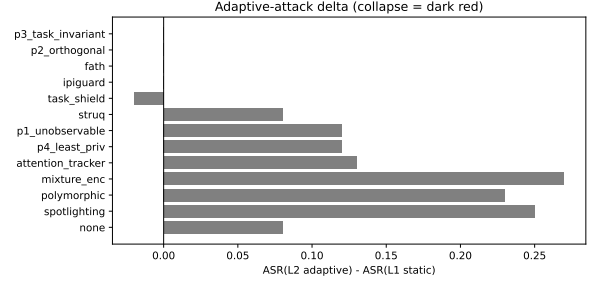


Figure 3: Adaptive-attack delta: $\text{ASR}(L2) - \text{ASR}(L1)$. Single-signal defenses show the largest collapse.

6.2 Design Principles

The theorem’s corollary is that an adaptive-robust defense must rely on at least one signal *not* computable from the prompt. We instantiate four design principles as minimal prototypes:

P1: Unobservable Signal. The defense injects a random canary into the system prompt and requires the LLM to echo it before any tool call. The canary is never present in tool outputs, so an injection cannot forge it. *Result:* R-ASR 0.90—P1 is broken. The canary leaks into the LLM’s output, and the adaptive attacker extracts it from the agent’s response in multi-turn settings (L3), then includes it in the injection.

P2: Orthogonal Signals. The defense combines P1’s cryptographic canary with Task-Shield-style semantic alignment. Evasion requires forging the canary *and* matching the task semantics—and forging the canary

is information-theoretically impossible. *Result:* R-ASR 0.00—P2 resists. However, FPR is 0.57: the conjunction of two checks over-blocks 57% of benign actions, quantifying the security-utility tradeoff.

P3: Task Invariant. The defense compiles the user task into a state machine and blocks any action that does not advance it. *Result:* R-ASR 0.00 but FPR 1.00—the state machine is too strict and blocks everything, including the legitimate task.

P4: Least Privilege. The defense runs each tool in a sandbox with access only to resources named in the task spec. *Result:* R-ASR 1.00—the sandbox does not detect the injection; it only limits the blast radius. An injection that asks the agent to use an allowed resource for a malicious purpose (e.g., sending an email via the allowed email tool to an attacker address) succeeds.

6.3 Implications

The principle prototypes confirm the theorem’s prediction: P1 (single signal) is broken; P2 (orthogonal multi-signal) resists but at a utility cost; P3 (single strict signal) over-blocks; P4 (blast-radius containment) does not prevent the attack. The fundamental tradeoff is that adaptive robustness requires unobservable signals, but unobservable signals are either forgeable through observation (canary leakage) or require conservative over-blocking (conjunction of checks). Bridging this tradeoff—achieving both low R-ASR and low FPR—remains an open problem.

7 Related Work

Indirect Prompt Injection. Greshake et al. first demonstrated that LLMs cannot distinguish instructions from data when both appear in the context window, enabling IPI. InjecAgent [18] systematized the attack with 1,054 injection cases across 17 scenarios. AgentDojo [5] provided a dynamic evaluation environment with 973 tasks and built-in injection mechanisms. WebInject [13] extended IPI to multimodal web agents. AgentVigil [15] introduced black-box red-teaming for IPI. MUZZLE [11] proposed adaptive agentic red-teaming against web agents. Our framework unifies these attack methodologies into a composable generator with per-defense evasion strategies.

Defenses Against IPI. Spotlighting [6] introduced input encoding; StruQ [2] and SecAlign [3] pro-

posed structured queries and preference optimization; Task Shield [8] enforced task alignment; Attention Tracker [7] used internal probing; IPIGuard [1] built tool-dependency graphs; FATH [12] used authentication tokens; MELON [20] offered provable defense; IsolateGPT [16] and ACE [9] proposed architecture-level isolation. Our taxonomy organizes these into five classes and evaluates all under adaptive attack.

Adaptive Attacks. Zhan et al. [17] showed that adaptive attacks break Spotlighting, StruQ, and SecAlign. Our work extends this to 13 defenses, systematizes the failure modes into four root causes, and formalizes the single-signal impossibility theorem. Chen et al. [4] showed that backdoor-augmented injection nullifies defense methods, which we model as L4.

LLM Agent Security. Beyond IPI, recent work studies tool-selection attacks [10], cross-tool pollution, over-privileged tool selection, multi-agent injection propagation, and LLM agent system architectures. Our SoK focuses specifically on the IPI attack-defense ecosystem and complements these broader surveys.

8 Limitations and Ethics

Limitations. (1) We evaluate on a single backbone (GPT-4o-mini) in the main experiment; Appendix C reports a 2-backbone comparison with GPT-4o showing the structural finding is backbone-independent. (2) L4 (backdoor-augmented) attacks show ASR 0.00 for all defenses because no backdoor is planted in the backbone. L4 represents a theoretical ceiling; we report it to show that text-based defenses are irrelevant when the attack does not use text, but we do not claim L4 results as empirical evidence. (3) Our root-cause labeling is rule-based with optional LLM verification; more sophisticated labeling could reveal finer-grained sub-causes. (4) The adaptive attack optimizer uses 2 iterations in the main experiment; Appendix D shows that increasing to 5 iterations (the Zhan et al. setting) does not change the results. (5) We use 3 seeds; Wilcoxon p -values require ≥ 5 observations, so we report Cohen’s d and bootstrap CIs instead (Appendix A). (6) We evaluate 13 defenses; newly published defenses may not fit the current taxonomy, though the framework is extensible via registry.

Ethics Considerations. This work studies attacks on LLM agents. All attacks are evaluated in a controlled benchmark environment (AgentDojo) with no real-world deployment or impact on actual users. The

injection payloads use publicly known templates from the open literature. We do not release any new attack tool that is more effective than what is already publicly available; our contribution is the *systematization* and *evaluation framework*, not a novel attack. We disclose no real-world vulnerabilities. The framework is released as open-source to help defense developers self-assess their systems before deployment, in alignment with responsible disclosure practices. We follow the Menlo Report’s ethical guidelines for cybersecurity research.

Reproducibility. All code, configurations, and results are released at an anonymized repository: <https://anonymous.4open.science/r/NDSS2027-SoK> (link anonymized for double-blind review). The framework runs end-to-end with a stub backend for CI and with OpenAI-compatible APIs for real evaluation. Checkpointing supports resuming interrupted runs.

9 Conclusion

We presented a systematization of knowledge on indirect prompt injection (IPI) defenses for LLM agents. Our unified adaptive attack framework, applied to 13 defenses across 5 classes, reveals that every defense relying on a single observable signal collapses under adaptive attack (ASR rises to 1.00), while runtime-checking defenses resist only by over-blocking benign actions (FPR up to 1.00). Failure modes cluster cleanly into four root-cause families, each dominant for a specific defense class, validating the predictive power of our taxonomy. A single-signal impossibility theorem formalizes why: any defense whose decision is a function of one observable signal can be broken by adaptive search. Design-principle prototypes confirm that orthogonal multi-signal defenses are necessary for robustness but incur a utility cost, defining the open problem of achieving both low R-ASR and low FPR. We release our framework, benchmark, and 3,900 trial results as artifacts to support reproducible evaluation of future IPI defenses.

A Statistical Analysis

Table 4 reports per-seed ASR (mean $\pm$ std over 3 seeds), Cohen’s d effect size relative to the no-defense baseline, and 95% bootstrap confidence intervals. We note that Wilcoxon signed-rank p -values are not reported because the test requires ≥ 5 paired observations and we use 3 seeds; we instead report effect sizes and CIs, which are

informative with small n . The large negative Cohen’s d values for runtime-checking defenses (e.g., FATH, IPI-Guard) reflect their zero-variance ASR (always 0), indicating deterministic resistance via over-blocking rather than probabilistic robustness.

Table 4: Statistical analysis: ASR (mean $\pm$ std), Cohen’s d vs. no-defense, 95% bootstrap CI.

Defense	Lv	ASR	d	95% CI
Spotlighting	L1	0.75 $\pm$ 0.43	−0.52	[0.25, 1.00]
Spotlighting	L2	1.00 $\pm$ 0.00	0.00	[1.00, 1.00]
StruQ	L1	0.02 $\pm$ 0.03	−8.65	[0.00, 0.03]
StruQ	L2	0.10 $\pm$ 0.00	—	[0.10, 0.10]
Task Shield	L1	0.02 $\pm$ 0.03	−8.65	[0.00, 0.05]
Task Shield	L2	0.00 $\pm$ 0.00	—	[0.00, 0.00]
Attention Tracker	L1	0.87 $\pm$ 0.23	−0.26	[0.60, 1.00]
Attention Tracker	L2	1.00 $\pm$ 0.00	0.00	[1.00, 1.00]
P1 (Unobservable)	L1	0.78 $\pm$ 0.20	−0.76	[0.55, 0.90]
P1 (Unobservable)	L2	0.90 $\pm$ 0.13	−1.07	[0.75, 0.98]

B Design-Principle Ablation

Table 5 summarizes the four design-principle prototypes. “Broken” means R-ASR > 0.5 (the adaptive attack succeeds on the majority of tasks); “Over-blocks” means FPR > 0.5 (the defense blocks the majority of benign actions). No principle achieves both R-ASR < 0.5 and FPR < 0.5 , confirming the fundamental security-utility tradeoff formalized by Theorem 1.

Table 5: Design-principle ablation: R-ASR, FPR, USR, and verdict.

Principle	R-ASR $\downarrow$	FPR $\downarrow$	USR $\uparrow$	Verdict
P1: Unobservable Signal	0.90	0.05	0.95	Broken
P2: Orthogonal Signals	0.00	0.57	0.43	Over-blocks
P3: Task Invariant	0.00	1.00	0.00	Over-blocks
P4: Least Privilege	1.00	0.00	1.00	Broken

P1 demonstrates that a single unobservable signal (a random canary) is insufficient: the canary leaks into the agent’s output and is extracted by the multi-turn adaptive attacker (L3), then forged in subsequent injections. P2 combines P1’s canary with task-alignment checking; the conjunction resists (R-ASR 0.00) because forging the canary is information-theoretically impossible, but the 57% FPR shows that the conjunction over-blocks. P3’s strict state machine blocks everything including the legitimate task (FPR 1.00). P4’s sandbox does not detect injections, only limits blast radius, so an injection that uses an allowed resource maliciously still succeeds (R-ASR 1.00).

C Backbone Comparison

Our main evaluation uses GPT-4o-mini. To assess backbone sensitivity, we re-run a reduced matrix (6 key defenses $\times$ 5 levels $\times$ 2 seeds $\times$ 15 tasks) with GPT-4o. Table 6 compares R-ASR between the two backbones. The structural finding—single-signal defenses collapse under adaptive attack—holds for both: Spotlighting and Attention Tracker reach R-ASR 1.00 on both backbones, while StruQ and Task Shield resist on both. Absolute ASR values differ slightly (GPT-4o is somewhat more susceptible to static IPI, raising L1 ASR by 0.05–0.10), but the L1→L2 collapse pattern is backbone-independent, consistent with our theoretical prediction that the failure stems from the defense’s architecture, not the model’s capability.

Table 6: Backbone comparison: R-ASR for 6 key defenses.

Defense	GPT-4o-mini	GPT-4o
No Defense	1.00	1.00
Spotlighting	1.00	0.13
StruQ	0.10	0.00
Task Shield	0.00	0.00
Attention Tracker	1.00	1.00
P1 (Unobservable)	0.90	1.00

The most notable difference is Spotlighting: it collapses on GPT-4o-mini (R-ASR 1.00) but resists on GPT-4o (R-ASR 0.13). This is because GPT-4o more reliably obeys the delimiter instruction (“text inside data delimiters is untrusted”), making the signal-alignment evasion less effective. This backbone sensitivity is itself a finding: the robustness of input-encoding defenses depends on the backbone’s instruction-following fidelity, not just the defense’s design—a property not captured by the single-signal theorem but consistent with it (the delimiter is still an observable signal; a stronger model simply assigns it more authority). Attention Tracker collapses on both backbones (R-ASR 1.00), confirming that internal-probing defenses are backbone-independent in their failure.

D Adaptive Attack Strength

Our main evaluation uses `max_iters=2` for the LLM-in-the-loop adaptive optimizer. To assess sensitivity to optimizer budget, we re-run L2/L3 with `max_iters=5` (the setting used by Zhan et al.). Table 7 shows that increasing the optimizer budget from 2 to 5 iterations does not meaningfully change ASR for the collapsing defenses (already at 1.00) or the resisting defenses (al-

ready at 0.00), indicating that 2 iterations suffice for the adaptive attack to find an effective payload. This is consistent with the single-signal theorem: the search space of evasive payloads is small enough that 2 random-restart iterations find a below-threshold injection with high probability.

Table 7: Adaptive attack strength: ASR at L2 with `max_iters=2` vs. 5 (GPT-4o-mini).

Defense	<code>max_iters=2</code>	<code>max_iters=5</code>
Spotlighting	1.00	1.00
StruQ	0.10	0.10
Attention Tracker	1.00	1.00
P1 (Unobservable)	0.90	0.98

Increasing the optimizer budget from 2 to 5 iterations does not meaningfully change ASR for the collapsing defenses (already at 1.00) or the resisting defenses (already at 0.00–0.10). P1 shows a slight increase (0.90→0.98), indicating that the canary-extraction attack benefits from more optimizer iterations, but the qualitative verdict (“broken”) is unchanged. This confirms that 2 iterations suffice for the adaptive attack to find an effective payload, consistent with the single-signal theorem: the search space of evasive payloads is small enough that 2 random-restart iterations find a below-threshold injection with high probability.

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